CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in agentic workflows and AI agents built on LLMs, from multi-step reasoning pipelines and RAG with vector databases to prompt engineering, evaluation and cloud deployment on AWS and Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, working directly with the CTO and SAP domain experts, backed by a strong foundation in machine learning and Python development across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Designed and optimized multi-agent extraction workflows with LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) for multi-step reasoning over complex regulatory and technical documents, raising precision/recall from 33%/45% to 93%/90% against an SAP-system-derived ground truth.
- Sole developer owning the full LLM knowledge-extraction pipeline and its integrations end-to-end: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code so AI agents can query it in agentic development workflows.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- Delivered as an AI-augmented engineer, using Claude Code, custom skills and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, RAG, GraphRAG, Vector Search, Neo4j, MongoDB, Postgres, FastAPI, React, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an AI agent with LangGraph that routes each request across vector database queries, general LLM inference and summarization, after mapping the decision-making workflows of business stakeholders.
- Developed a React front-end and a Python backend with FastAPI, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Designed Retrieval-Augmented Generation (RAG) pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- Automated RAG generation and retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.

- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- **Tech stack: Python, LangChain, Hugging Face Transformers, PyTorch, RAGAS, PytrecEval, Vector Databases, Azure AI Search, AWS (Lambda, SQS, DynamoDB, CDK), Docker, Streamlit**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Meme Generation with LLMs | Open Llama 3B | QLoRA | Hugging Face
- Fine-tuned the open-source Open Llama 3B model with QLoRA to generate memes, covering dataset preparation, parameter-efficient fine-tuning and prompt-based generation.

TECHNICAL SKILLSET

- **LLMs & Agents:** LLMs (Claude via Claude Code/MCP, Llama, open-source LLMs), LangGraph (agentic workflows, multi-agent), LangChain, LiteLLM, MCP, prompt engineering, QLoRA fine-tuning
- **RAG & Retrieval:** RAG, GraphRAG, Vector Databases (Qdrant, Pinecone, Azure AI Search), embeddings, RAG evaluation (RAGAS, PytrecEval)
- **Programming & Frameworks:** Python, FastAPI, React (beginner), Postman, Scikit-Learn, PyTorch, TensorFlow, Hugging Face Transformers, NLTK
- **Databases:** SQL, Postgres, MongoDB, Neo4j, SPARQL
- **Cloud & DevOps:** Docker, Kubernetes, Azure, AWS (S3, EC2, Lambda, SQS, DynamoDB, VPC, CDK), Git, GitLab CI/CD
- **Methodologies:** Agile, Claude Code / agentic coding
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI